- ***1*** Sets the update frequency for synchronizing the target model parameters to the main DQN
- ***2*** Uses the target network to get the maximum Q value for the next state
- ***3*** Copies the main model parameters to the target network

When we plot the loss for a target network approach with experience replay (figure 3.16), we still get a noisy loss plot, but it is significantly less noisy and clearly down-trending. You should try to experiment with the hyperparameters, such as the experience replay buffer size, the batch size, the target network update frequency, and the learning rate. The performance can be quite sensitive to these hyperparameters.

Figure 3.16. The DQN loss plot after including a target network to stabilize training. This shows a much faster training convergence than without the target network, but it has noticeable spikes of error when the target network synchronizes with the main DQN.

